

PGCP-FBD-2026

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Course Name: Devops and challenges in Fintech and blockchain

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Q1 What is the git command to initialize the new Git repository ?

Ans: git init

```
~/Desktop/GITLAB/GitAssignment/git --zsh
himanshukamble@Himanshus-MacBook-Air Desktop % cd GITLAB
himanshukamble@Himanshus-MacBook-Air GITLAB % ls
himanshukamble@Himanshus-MacBook-Air GITLAB % mkdir GitAssignment
himanshukamble@Himanshus-MacBook-Air GITLAB % ls
GitAssignment
himanshukamble@Himanshus-MacBook-Air GITLAB % cd GitAssignment
himanshukamble@Himanshus-MacBook-Air GitAssignment % ls -la
.
..
himanshukamble@Himanshus-MacBook-Air GitAssignment % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/himanshukamble/Desktop/GITLAB/GitAssignment/.git/
himanshukamble@Himanshus-MacBook-Air GitAssignment % ls -la
.
..
.git
himanshukamble@Himanshus-MacBook-Air GitAssignment % cd .git
himanshukamble@Himanshus-MacBook-Air .git % ls
HEAD      config      description  hooks      info      objects     refs
himanshukamble@Himanshus-MacBook-Air .git %
```

Q2 What is the command to add file to the staging area

Ans: git add . git add <filename>

```
himanshukamble@Himanshus-MacBook-Air GitAssignment % echo "Hello" > file.txt
himanshukamble@Himanshus-MacBook-Air GitAssignment % ls
file.txt
himanshukamble@Himanshus-MacBook-Air GitAssignment % git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file.txt

nothing added to commit but untracked files present (use "git add" to track)
himanshukamble@Himanshus-MacBook-Air GitAssignment % git add .
himanshukamble@Himanshus-MacBook-Air GitAssignment % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file.txt

himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q3) What is the command to commit changes with the message ?

Ans: `git commit -m "File is added"`

```
~/Desktop/GITLAB/GitAssignment -- zsh
himanshukamble@Himanshus-MacBook-Air GitAssignment % echo "Hello" > file.txt
himanshukamble@Himanshus-MacBook-Air GitAssignment % ls
file.txt
himanshukamble@Himanshus-MacBook-Air GitAssignment % git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file.txt

nothing added to commit but untracked files present (use "git add" to track)
himanshukamble@Himanshus-MacBook-Air GitAssignment % git add .
himanshukamble@Himanshus-MacBook-Air GitAssignment % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   file.txt

himanshukamble@Himanshus-MacBook-Air GitAssignment % git commit -m "File is added"
[master (root-commit) fb9792f] File is added
Committer: Himanshu Kamble <himanshukamble@Himanshus-MacBook-Air.local>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+)
create mode 100644 file.txt
himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q4) What is the command to check the status of a repository ?

Ans: `git status`

```
~/Desktop/GITLAB/GitAssignment -- zsh
himanshukamble@Himanshus-MacBook-Air GitAssignment % git status
On branch master

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file2.txt

nothing added to commit but untracked files present (use "git add" to track)
himanshukamble@Himanshus-MacBook-Air GitAssignment % git add .
himanshukamble@Himanshus-MacBook-Air GitAssignment % git status
On branch master

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   file2.txt

himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q5) What is the command to create a new branch and switch to it.

Ans:

```
~/Desktop/GITLAB/GitAssignment -- zsh
himanshukamble@Himanshus-MacBook-Air GitAssignment % git checkout -b second
Switched to a new branch 'second'
himanshukamble@Himanshus-MacBook-Air GitAssignment % git switch -c second
fatal: a branch named 'second' already exists
himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q6) What is the command to merge the branch into the current branch?

Ans:

```
~/Desktop/GITLAB/GitAssignment -- zsh
himanshukamble@Himanshus-MacBook-Air GitAssignment % git checkout -b second
Switched to a new branch 'second'
himanshukamble@Himanshus-MacBook-Air GitAssignment % git switch -c second
fatal: a branch named 'second' already exists
himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q7 What is the command to view the difference between two branches ?

Ans:

git diff master

git diff second

```
himanshukamble@Himanshus-MacBook-Air GitAssignment % git diff master second
diff --git a/file2.txt b/file2.txt
new file mode 100644
index 0000000..e69de29
himanshukamble@Himanshus-MacBook-Air GitAssignment % git diff second
diff --git a/file2.txt b/file2.txt
new file mode 100644
index 0000000..e69de29
himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q8 What is the command to revert changes to the specific file. ?

Ans :

```
himanshukamble@Himanshus-MacBook-Air GitAssignment % git restore --source fb9792f file.txt
himanshukamble@Himanshus-MacBook-Air GitAssignment %
```

Q9) What is the command to undo the last commit ?

Ans: git reset --soft HEAD~1

git reset --hard HEAD~1

```
~/Desktop/GITLAB/GitAssignment -- zsh
himanshukamble@Himanshus-MacBook-Air GitAssignment % git reset --soft HEAD~1
fatal: ambiguous argument 'HEAD~1': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
himanshukamble@Himanshus-MacBook-Air GitAssignment % git reset --hard HEAD~1
fatal: ambiguous argument 'HEAD~1': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
```

Q10) What is the command to push changes to a remote repository ?

Ans

git push origin <branch name>

```
himanshukamble@Himanshus-MacBook-Air GitAssignment % git remote add origin https://github.com/hkamble11/GITAssignment.git
himanshukamble@Himanshus-MacBook-Air GitAssignment % git remote -v
origin  https://github.com/hkamble11/GITAssignment.git (fetch)
origin  https://github.com/hkamble11/GITAssignment.git (push)
himanshukamble@Himanshus-MacBook-Air GitAssignment % git push origin second
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 235 bytes | 235.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hkamble11/GITAssignment.git
 * [new branch]      second -> second
himanshukamble@Himanshus-MacBook-Air GitAssignment %
```